

ANLP HW4 Final Report

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Abstract

Hypothetical Document Embeddings (HyDE) improves zero-shot dense retrieval by using an LLM-generated hypothetical answer as the retrieval probe, but unconstrained generation can introduce hallucinated or tangential details that push the hypothetical embedding away from the relevant cluster: a failure mode we observed in our baseline reproduction and call *embedding drift*. We propose **KG-Grounded HyDE**, which constrains hypothetical generation with entity–relation context retrieved from a knowledge graph built offline over the corpus, keeping the pipeline training-free. On SciFact, KG-Grounded HyDE improves over vanilla HyDE on every metric (Recall@10 +1.3, MRR@10 +3.2, nDCG@10 +2.5) and recovers the ranking precision that vanilla HyDE sacrifices relative to dense retrieval. The gains from hypothetical retrieval hinge less on generating more and more on generating under the right constraints. Our code is available on GitHub¹.

1 Introduction

Dense retrieval underpins most retrieval-augmented generation (RAG) systems, but it carries a persistent failure mode: a user’s query and the document that answers it often do not share enough vocabulary or phrasing to land in the same neighborhood of embedding space (Lewis et al., 2020). Short, underspecified questions are especially vulnerable, and a retriever that misses the relevant evidence cannot be rescued by a stronger generator downstream.

Hypothetical Document Embeddings (HyDE) addresses this gap by prompting an instruction-following LLM to write a hypothetical answer to the query and embedding that synthetic text as the retrieval probe (Gao et al., 2023). Even when the hypothetical contains factual errors, its embedding tends to land closer to the relevant document cluster

than the raw query does, and HyDE has become a standard tool for zero-shot dense retrieval across BEIR, web search, and multilingual benchmarks.

Free-form generation, however, introduces a new problem: with no external grounding, the LLM has wide latitude to inject tangential or hallucinated content. Our reproduction of HyDE on SciFact surfaced this concretely. While HyDE improved Recall@10 over a strong BGE dense baseline, it *lowered* MRR@10 from 69.4 to 69.0, with topically relevant but tangentially detailed hypotheticals pushing the gold document down the ranked list. The HyDE paper itself acknowledges that the generated document may be “ungrounded factually” (Gao et al., 2023); we show that this ungroundedness has a measurable cost in ranking precision.

Research gap. Existing extensions to HyDE largely operate at the indexing stage. HyPE precomputes hypothetical questions over the corpus (Vake et al., 2025), and TreeQA decomposes queries into sub-question/hypothetical-answer pairs for multi-hop reasoning (Zhang et al., 2025). Graph-based retrievers such as GraphRAG (Edge et al., 2025) use knowledge graphs to organize the corpus itself for global, sense-making queries. None of these methods directly target the per-query drift that arises when a single hypothetical document is generated freely from the LLM’s parametric knowledge.

Our approach. We propose **KG-Grounded HyDE**, which inserts a structured grounding step between the query and the hypothetical generator. Offline, we extract entity–relation (E–R) pairs from each corpus document using an LLM and store them in a queryable graph. At query time, we extract key entities from the query, retrieve the most relevant E–R pairs from the graph, and prompt the LLM to generate a hypothetical answer *conditioned on those E–R pairs*. The grounded hypothetical is then embedded and retrieved exactly as in standard HyDE. Crucially, our use of the knowledge graph

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is targeted rather than global: we never summarize the corpus or materialize community-level structures, so the method remains training-free and adds only one short LLM call and one graph lookup per query.

Results and contributions. On SciFact, KG-Grounded HyDE outperforms vanilla HyDE on Recall@10 (88.2 $\rightarrow$ 89.5), Recall@100 (96.7 $\rightarrow$ 97.3), MRR@10 (69.0 $\rightarrow$ 72.2), and nDCG@10 (73.4 $\rightarrow$ 75.9). The MRR@10 result is particularly informative: vanilla HyDE traded 0.4 points of ranking precision for its recall gains, while KG-Grounded HyDE recovers that loss and adds 2.8 points on top, indicating that structured grounding stabilizes the hypothetical without sacrificing its expansion benefits. A residual error analysis shows that the failures KG grounding cannot fix tend to involve quantitative claims (where the qualitative E-R structure provides little leverage) or queries whose granularity does not match the corpus, suggesting natural extensions for future work. More broadly, our findings argue that the next gains in query-side LLM augmentation will come not from generating richer or longer hypotheticals, but from generating them under the right structural constraints.

2 Related Work

2.1 Hierarchical Indices

Traditional RAG (Lewis et al., 2020) relies on flat chunking (e.g., splitting documents into fixed-size token blocks). While simple to implement, flat retrieval suffers from *semantic fragmentation*: if an answer spans multiple chunks or requires document-level understanding, the retriever is forced to choose between shallow matches and broader context. The field has responded by shifting toward hierarchical RAG, which organizes knowledge into multi-level trees or graphs to support multi-resolution retrieval.

RAPTOR (Sarathi et al., 2024) is a representative hierarchical approach: it clusters text chunks and recursively summarizes each cluster with an LLM, producing a bottom-up tree where queries can be matched against both high-level summaries and low-level leaf chunks at retrieval time. This allows the retriever to dynamically choose its granularity per query rather than committing to a fixed chunk size.

2.2 Knowledge Graphs for Retrieval

A parallel line of work uses knowledge graphs to encode corpus-level structure that flat embeddings cannot easily capture. GraphRAG (Edge et al., 2025) converts the corpus into a hierarchical knowledge graph by extracting entities and relations from each chunk (using LLMs in the manner of (Ban et al., 2023)) and merging them into a global graph. Hierarchical community detection then groups strongly connected entities into multi-level communities, each of which is summarized by the LLM. At query time, GraphRAG uses a map-reduce process: community summaries generate partial answers that are merged into a final response. This makes GraphRAG well suited to global, sense-making questions (Klein et al., 2006) that require synthesizing information across the entire corpus.

The trade-off, however, is that GraphRAG-style methods are heavyweight (they materialize and summarize the full corpus graph) and are optimized for whole-corpus synthesis rather than narrow fact retrieval, where vector RAG remains stronger.

2.3 Hypothetical Representations

Hypothetical-representation methods address a complementary problem: the user query and the target chunk often live in different semantic forms, with the short query lacking the lexical or embedding-level overlap needed to surface the answering passage. The shared intuition across this literature is that an LLM-generated intermediate representation, i.e., a hypothetical document, answer, question, or prompt, can better expose the latent information need and align query and corpus more effectively than the raw query alone.

The most influential work in this space is HyDE (Gao et al., 2023), which prompts an instruction-following LLM to generate a hypothetical document for a query, embeds it, and uses the embedding to retrieve real corpus documents. HyDE achieves strong zero-shot retrieval on BEIR, TREC DL, and Mr. TyDi without any relevance labels.

HyPE (Vake et al., 2025) moves the hypothetical generation step from query time to indexing time: for each chunk, it precomputes multiple hypothetical questions and stores their embeddings, reframing retrieval as question-to-question matching. This trades larger index size for lower per-query latency and reflects a broader trend toward offline, corpus-side hypothetical augmentation.

TreeQA (Zhang et al., 2025) uses hypothetical answers inside a structured multi-hop reasoning pipeline rather than as a one-shot retrieval probe. Each sub-question is paired with a verifiable hypothetical answer that supports decomposition and intermediate verification, making the hypothetical lighter and more focused than HyDE’s full document.

2.4 Trends and Shortcomings

Three patterns emerge across this prior work. First, the field is moving from simple lexical query expansion toward *representation engineering*, where generated text actively reshapes the semantic space between queries and documents. Second, there is a clear shift from online per-query generation toward offline indexing-time generation, motivated by latency. Third, researchers increasingly recognize that purely parametric hypotheticals, drawn from the LLM alone, are unreliable, and corpus-grounded hypotheticals are gaining traction.

What remains underexplored is the intersection of these trends at *query time*. HyDE is fully online and fully ungrounded; HyPE is offline; GraphRAG uses graphs but operates over the full corpus; and TreeQA constrains hypotheticals through tree decomposition rather than through structured corpus knowledge. None of these methods directly addresses the drift that arises when a single, free-form hypothetical document is generated for a single query, which our error analysis identified as a measurable source of ranking-precision loss in vanilla HyDE. Our work targets this specific gap by introducing per-query knowledge-graph grounding into the HyDE pipeline, retaining its zero-shot, online character while constraining what the LLM is allowed to write.

3 Data

3.1 SciFact

We evaluate KG-Grounded HyDE on **SciFact** (Wadden et al., 2020), a scientific fact-verification dataset distributed as part of the BEIR benchmark (Thakur et al., 2021). We load the dataset via the mteb/scifact Hugging Face mirror, which preserves the original BEIR formatting. SciFact pairs short scientific claims with abstracts from a biomedical-and-life-sciences corpus and asks systems to retrieve the abstract that supports or refutes each claim. Our indexed corpus contains **5,183 abstracts**, and we evaluate on the **300 test claims**

that remain after filtering qrels with non-positive relevance scores and restricting to claims whose gold documents are present in the corpus. Each claim is associated with one or more annotated gold abstracts, and queries with multiple gold documents are evaluated against the full set of relevant abstracts.

Why SciFact? SciFact is well suited to our research question for three reasons. First, it is the dataset on which we observed the most pronounced embedding-drift behavior during baseline reproduction (MRR@10 declining from 69.4 to 69.0 despite Recall@10 gains), making it the natural testbed for any method that claims to address drift. Second, its biomedical domain is dense in named entities and entity–relation structure (genes, proteins, diseases, treatments, pathways), which a knowledge graph can capture well; this is the regime where structured grounding has the most leverage. Third, SciFact’s claims are typically short (mean 12.5 words, median 12 words across the 300 evaluated queries), which is precisely the query profile where HyDE’s vocabulary-bridging behavior matters most.

Preprocessing. Each document is indexed as a single string of the form “Title: {title}\nPassage: {text}”, with no chunking, lowercasing, stemming, or other normalization applied. This whole-abstract representation is consistent with both the original HyDE evaluation (Gao et al., 2023) and standard BEIR practice for SciFact, where abstracts are short enough to embed in a single pass. Claims are likewise used verbatim. The only filtering we apply is on the query side: we drop qrels with score ≤ 0 and exclude any test claim whose gold document is missing from the corpus, leaving 300 evaluable queries. Knowledge-graph construction (entity–relation extraction over the SciFact corpus) is a separate offline pipeline described in Section 4 (Method) rather than treated as data preprocessing, since it produces a derived structure rather than modifying the underlying retrieval corpus.

3.2 TREC DL19 (Baseline Reproduction Only)

For our initial baseline reproduction (Section 4.1), we additionally evaluated on **TREC DL19** (Craswell et al., 2020), the passage-retrieval task built over the MS MARCO v1 passage corpus (Nguyen et al., 2016). DL19 provides 43 test queries with deep relevance judgments over an $\approx 8.8\text{M}$ -passage corpus, and it is one of the two

settings (alongside BEIR) on which the original HyDE paper reports its primary results. We included DL19 to verify that our reproduction recovers the trends in Gao et al. (2023) on a different domain (web passages versus biomedical abstracts) and at a substantially larger corpus scale. DL19 was not used for evaluating our proposed method; all experiments comparing dense retrieval, vanilla HyDE, and KG-Grounded HyDE are conducted on SciFact, where our error analysis identified the drift behavior we set out to fix.

4 Method

4.1 Baseline Reproduction

We reproduce the method proposed in Precise Zero-Shot Dense Retrieval without Relevance Labels (Gao et al., 2023), which introduces Hypothetical Document Embeddings (HyDE) for zero-shot dense retrieval. The task we focus on is open-domain document retrieval, where the goal is to retrieve relevant documents given a natural language query without supervised relevance labels. This paper is chosen because HyDE represents a strong and widely adopted baseline for zero-shot retrieval, and it directly addresses the semantic gap between queries and documents by leveraging LLMs to generate intermediate representations. Moreover, it provides a suitable foundation for our later improvements on grounding and stability.

Our implementation closely follows the original HyDE pipeline, consisting of four stages: (1) generating a hypothetical document from the query using an LLM, (2) embedding the generated text, (3) performing vector similarity search. Specifically, we use BGE-M3 (and BGE-base-en-v1.5 for controlled comparison) as the embedding model, and FAISS as the vector search backend. Compared to the original paper, we replace the deprecated text-davinci-003 with gpt-4o-mini (or Qwen-based models) to ensure reproducibility with modern APIs, while keeping the rest of the pipeline consistent. This substitution is justified as newer models achieve comparable or stronger performance with lower cost and better availability.

We conduct experiments on two standard BEIR benchmark datasets: DL19 and SciFact. DL19 is used to verify consistency with the original paper, while SciFact provides a more controlled setting to analyze the interaction between HyDE and different embedding models. Additionally, we replace the original Contriever encoder with BGE-base-en-

v1.5 to study how stronger embedding representations affect the marginal benefit of HyDE. This design allows us to isolate the contribution of the generation module and evaluate whether HyDE still provides gains when the base encoder is already strong.

4.2 Proposed methodology

Our proposed method extends HyDE by introducing knowledge graph (KG)-grounded hypothetical representations to improve retrieval stability. The architecture follows a modified HyDE pipeline: given a query, we first extract key entities and concepts, then retrieve related entity–relation (E-R) pairs from a pre-built KG constructed offline from the corpus. These structured signals are used to constrain the LLM during generation, prompting it to produce a grounded hypothetical document instead of an unconstrained free-form output. This design directly addresses the embedding drift issue in vanilla HyDE, where hallucinated or tangential details can mislead retrieval.

The generated hypothetical document is then encoded using a dense embedding model (e.g., BGE) and used for vector retrieval, keeping the rest of the pipeline consistent with standard HyDE. The method does not require additional model training, relying instead on modular components including entity extraction, KG retrieval, and prompt-based generation.

The key modification is KG grounding, which injects explicit relational structure into the generation process, ensuring that the hypothetical document remains semantically aligned with relevant concepts in the corpus. By anchoring generation to entity–relation evidence, our approach produces more accurate embeddings, improving both retrieval robustness and ranking quality.

4.3 Rationale

Our method addresses a key limitation of HyDE: embedding drift caused by unconstrained generation. While HyDE improves recall by expanding queries, the LLM may introduce irrelevant, tangential or hallucinated details, shifting embeddings away from truly relevant documents and hurting ranking quality.

We mitigate this by grounding generation in a knowledge graph (KG). Specifically, we retrieve entity–relation pairs related to the query and use them to constrain the LLM’s hypothetical document generation. This reduces hallucination and keeps the

generated representation semantically aligned with the corpus. Compared to prior work, our approach preserves HyDE’s query expansion benefits while improving stability and precision, without requiring additional training or heavy global graph processing.

4.4 Implementation Details

Our method is implemented as a hybrid GraphRAG–HyDE–dense retrieval pipeline. Knowledge graph construction is performed using GraphRAG (v3.0.9), integrating LLM-based entity–relation extraction and LanceDB for vector indexing. The SciFact corpus contains 5,183 documents and 300 queries, and is segmented into 512-token chunks (no overlap) prior to extraction. Empirical analysis shows that SciFact documents are relatively short (mean $\approx$ 320 tokens, p95 $\approx$ 542), with 93 % fitting into a single chunk. This motivates the use of 512-token chunking without overlap, which avoids redundancy. Overlapping chunks would duplicate content, inflate entity–relation counts, increase embedding cost, and introduce redundant evidence during retrieval.

The constructed knowledge graph contains 25,791 entities and 35,372 relationships, with all entities indexed in LanceDB (25,791 vectors). Graph construction is cache-based and parallelized (8 threads) under rate-limited API usage. We use OpenAI gpt-4o-mini for entity–relation extraction and hypothetical document generation, and text-embedding-3-large for graph-side embeddings. Community-level summarization is omitted to reduce overhead.

At query time, we first extract entities and relations from the input query, then use them to initialize graph retrieval by selecting 3 seed entities in the KG. From these seeds, we expand the graph to retrieve the top-10 related entities and top-10 relationships, which are injected into the prompt to constrain hypothetical document generation (temperature=0.3, max tokens=256). The generated representation is then embedded using BAAI/bge-base-en-v1.5 and used for dense retrieval over the original corpus via FAISS, following the HyDE framework.

The system is fully inference-based, with no training or fine-tuning, and is evaluated on the SciFact test split using standard retrieval metrics.

4.5 Experimental Design

We evaluated on SciFact (5,183 documents, 300 queries) by comparing three methods: dense retrieval (BGE), vanilla HyDE, and our KG-grounded HyDE. All methods share the same corpus and embedding model to isolate the effect of KG grounding.

We use standard retrieval metrics: Recall@k (k=10,100) to measure coverage, MRR@10 to evaluate ranking precision, and nDCG@10 for overall ranking quality. These metrics capture the trade-off in prior work where HyDE improves recall but may hurt ranking due to embedding drift. Our experiments test whether KG grounding can maintain recall gains while improving ranking stability and precision.

5 Results

5.1 Baseline Results

We first reproduced the HyDE baseline on the DL19 dataset to ensure our implementation follows the original retrieval pipeline and produces comparable results. The reproduced results closely match the originally reported performance, with only minor differences across all metrics (e.g., nDCG@10 of 61.0 vs. 61.3, MAP of 40.8 vs. 41.8). These small gaps are expected as our implementation replaces the deprecated text-davinci-003 with gpt-4o-mini for hypothetical document generation. The rest of the HyDE pipeline is kept consistent.

Table 1: Reproduction Results on DL19 Dataset

Method	nDCG@10	Recall@100
BM25	67.9	92.5
Contriever	64.9	92.6
HyDE	69.1	92.4

We then evaluate our baselines on the SciFact dataset, comparing BM25, dense retrieval, and HyDE-based retrieval using both Contriever and BGE-base-en-v1.5 as the embedding encoder. BGE-base-en-v1.5 (109M) is a same-sized alternative to Contriever (110M) that benefits from improved training techniques, allowing a fair architectural comparison while acknowledging that BGE’s stronger baseline performance may reduce the marginal impact of HyDE. The results confirm this expectation. With Contriever, HyDE produces a large improvement, raising Recall@10 from 41.8

to 78.7. With the stronger BGE encoder, the improvement is smaller: Recall@10 increases from 85.7 to 88.2, and Recall@100 increases from 96.3 to 96.7. However, BGE-HyDE slightly decreases MRR@10 from 69.4 to 69.0, suggesting that HyDE can improve retrieval coverage while sometimes hurting the exact ranking of the relevant document.

Table 2: Baseline Reproduction on SciFact Dataset

Method	Recall@10	Recall@100	MRR@10	nDCG@10
BM25	80.7	92.5	64.6	68.0
BGE_dense	85.7	96.3	69.4	73.0
BGE_HyDE	88.2	96.7	69.0	73.4
Contriever_dense	41.8	90.3	54.5	58.5
Contriever_HyDE	78.7	94.5	56.6	61.1

To better understand HyDE’s behavior, we conduct a per-query hit@10 analysis comparing BGE dense retrieval and BGE-HyDE on the 300 SciFact test queries. These queries are categorized into 4 buckets: both hit, both miss, HyDE helped, and HyDE hurt. The results can be observed in Table 3.

Table 3: Per-query retrieval outcome categories (hit@10) on SciFact, comparing BGE dense retrieval vs. BGE + HyDE (GPT-4o-mini).

Category	Count	%
Both hit	252	84.0
Both miss	25	8.3
HyDE helped	15	5.0
HyDE hurt	8	2.7
Total	300	100.0

In the previous table, "both hit" means both methods retrieved the golden document in the top 10. "HyDE helped" means only HyDE found it, "HyDE hurt" means only BGE dense retrieval found it, and "both miss" means neither did. The interesting result here is that the 84% agreement rate shows the methods largely behave the same, the 5% HyDE-helped is almost double the 2.7% HyDE-hurt, and the 8.3% both-miss represents a ceiling that neither method can crack.

HyDE has a net positive effect, helping on 15 queries while hurting only 8. Among the HyDE-helped cases, queries tend to be shorter and lack scientific vocabulary, allowing the hypothetical document to bridge the semantic gap. Among the HyDE-hurt cases, we observe an embedding drift pattern, where the generated hypothetical document introduces tangential details that push the gold document out of the top 10 despite it remaining in the top 100. This embedding drift also explains why

HyDE improves Recall@10 but slightly decreases MRR@10. These 8 queries can be seen more in detail in Table 4.

Table 4: Queries where HyDE degraded retrieval (baseline hit@10, HyDE miss@10). Gold Doc Rank shows the position of the relevant document in each method’s ranked list.

QID	Claim (truncated)	Baseline Rank	HyDE Rank
99	Alizarin forms hydrogen bonds with PGAM1...	1	18
549	IRG1 has antiviral effects against neurotropic viruses	1	12
690	<10% of Gabonese children with SFM had...	2	60
1107	Subcutaneous fat depots undergo browning...	3	16
415	Female APOE4 carriers have increased dementia risk	4	13
715	Low miR7a expression represses target genes...	5	11
785	Microarray results from culture-amplified...	7	14
597	Incidence rates of cervical cancer have decreased	11	19

5.2 KG-Grounded HyDE Results

In our experiment, we evaluated the proposed KG-Grounded HyDE method on SciFact using the same BGE encoder and retrieval setup. The goal is to test whether knowledge-graph grounding can preserve HyDE’s recall benefit while reducing embedding drift and improving ranking quality.

Table 5 shows that KG-Grounded HyDE improves over vanilla HyDE on all evaluation metrics. Recall@10 increases from 88.2 to 89.5, and Recall@100 increases from 96.7 to 97.3, showing that structured grounding helps create more accurate query-side embeddings while preserving HyDE’s original query expansion benefit.

Table 5: Experiment Results on SciFact Dataset

Method	Recall@10	Recall@100	MRR@10	nDCG@10
BGE_dense	85.7	96.3	69.4	73.0
BGE_HyDE	88.2	96.7	69.0	73.4
BGE_KG_HyDE(Ours)	89.5	97.3	72.2	75.8

The improvement in MRR@10 is especially important. While vanilla HyDE lowers MRR@10 from 69.4 to 69.0 compared with BGE dense retrieval, KG-Grounded HyDE raises it to 72.2. This means vanilla HyDE sacrifices ranking precision for broader retrieval coverage, while KG-Grounded HyDE preserves both. This result supports our central hypothesis that KG grounding reduces embedding drift by constraining generation and keeping the gold document closer to the top of the ranked list.

The nDCG@10 result further confirms this pattern. The gain from BGE dense retrieval to vanilla HyDE is only 0.4 points, whereas the gain from vanilla HyDE to KG-Grounded HyDE is 2.5 points. This suggests that the primary source of improvement is not the length of the generated representa-

tion, but the degree to which it is constrained by structured knowledge.

5.3 Error Analysis

We analyze the 27 queries where KG-Grounded HyDE fails to retrieve the gold document in the top 10. These failures fall into two categories.

Semantic Generality Mismatch (20/27, 74%)

The dominant failure mode occurs when the query is phrased as a broad, general claim, but the gold document addresses a highly specific sub-problem within the same topic. In these cases, the KG retrieval works correctly and returns relevant entities, but the HyDE generation produces a textbook-level passage that fails to capture the specificity of the actual paper. For example, the query "Glycolysis is one of the primary glycometabolic pathways in cells" is a general statement, while the gold document studies a niche mechanism involving vesicular glycolysis in axonal transport. HyDE generates a generic description of the glycolysis pathway, which causes FAISS to rank many superficially similar glycolysis papers above the gold document. The root cause is the mismatch between the query's abstraction level and the paper's specificity. Although the KG provides topically correct context, it is not specific enough to steer HyDE toward the particular angle the paper takes.

Quantitative Mismatch (7/27, 26%) The remaining failures involve queries containing specific statistics or numerical claims. In these cases, the KG stores relational facts (e.g., that low birth weight is associated with perinatal mortality) but does not capture quantitative values. As a result, HyDE generates a qualitatively correct passage but omits or approximates the exact figure, causing the embedding to drift away from the gold document that is distinguished precisely by that statistic. This is a systematic limitation of the KG-grounded pipeline, as knowledge graphs are inherently qualitative structures that capture entities and relationships but strip out numerical details during extraction.

6 Discussion

6.1 Interpretation of Results

The results indicate that HyDE's primary contribution is query expansion. The generated hypothetical document introduces domain-specific vocabulary that the raw query lacks, helping the retriever surface relevant documents that would otherwise

be missed. This effect is particularly pronounced for short SciFact claims, which often contain insufficient specialized terminology for effective dense matching.

However, increasing the volume of generated text does not consistently improve retrieval quality. With the BGE encoder, vanilla HyDE improves Recall@10 but slightly lowers MRR@10 (69.4 $\rightarrow$ 69.0), indicating that the generated document can help retrieve the correct evidence but at the cost of placing it at a lower rank position. The error analysis attributes this to embedding drift. Tangential details in the hypothetical document shift the query embedding away from the gold document. This observation is consistent with the original HyDE paper's acknowledgment that hypothetical documents are "unreal" and may contain inaccurate details (Gao et al., 2023).

KG-Grounded HyDE addresses this instability by conditioning generation on entity-relation pairs retrieved from the knowledge graph. The improvements in MRR@10 and nDCG@10 suggest that structured grounding produces a more focused hypothetical document that is better aligned with the relevant corpus evidence. Our central finding is therefore that HyDE benefits more from constrained generation than from simply richer or longer generation.

At the same time, the error analysis reveals that grounding alone is insufficient. When the query and gold document operate at different levels of specificity, or when the query depends on an exact numerical value, the knowledge graph provides insufficient guidance for effective generation. These cases suggest that the effectiveness of KG-Grounded HyDE depends not only on the topical relevance of the retrieved graph context but also on the degree to which it captures the granularity and specificity of the target document.

6.2 Strengths of KG-Grounded HyDE

KG-Grounded HyDE preserves the recall benefit of query expansion while resolving the ranking instability identified in vanilla HyDE. This is practically important since downstream components in RAG pipelines typically consume only a small number of top-ranked passages. The method also remains entirely training-free, adding only an offline graph construction step and a lightweight query-time grounding step, which makes it modular and straightforward to integrate with existing dense retrieval pipelines.

6.3 Weakness of KG-Grounded HyDE

The primary weakness is the method’s dependence on the quality and coverage of the knowledge graph, which becomes a new source of error. As shown in the error analysis, both the semantic generality mismatch and quantitative mismatch failures stem from the knowledge graph encoding qualitative entity-relation triples that lack sufficient specificity for certain retrieval targets. A second weakness is the added system complexity, including offline entity extraction, graph storage, graph querying, and prompt construction.

6.4 Practical and Theoretical Implications

KG-Grounded HyDE is most suited for retrieval settings where queries are short, domain-specific, and entity-rich. Theoretically, the results support the view that query-side LLM augmentation should be controlled rather than open-ended. The failure analysis suggests that future work may benefit from preserving quantitative information during KG extraction and implementing a selective strategy that chooses among retrieval methods based on query characteristics.

7 Conclusion & Future Work

Our work addresses the instability of HyDE in dense retrieval, particularly the embedding drift caused by unconstrained LLM generation. Our error analysis shows that while HyDE improves recall for underspecified queries (5% helped vs. 2.7% hurt), it can degrade ranking by introducing tangential details, pushing relevant documents lower in the ranking. To address this, we propose KG-grounded HyDE, which constrains hypothetical document generation using entity–relation signals from a knowledge graph.

Experimental results demonstrate that this design leads to more stable and accurate retrieval, improving both ranking precision and overall quality. Compared to vanilla HyDE, our method achieves gains in MRR@10 (+3.2%) and nDCG@10 (+2.4%), while maintaining or slightly improving recall. These results confirm that grounding generation with structured knowledge effectively mitigates drift and produces embeddings that better align with relevant documents.

For future work, several directions are worth exploring. First, instead of rebuilding the knowledge graph from scratch, we can study incremental KG construction, enabling efficient updates

when new documents are added. Second, since entity–relation extraction and hypothetical document generation are critical components, improving system prompts and prompt design may further enhance performance. Third, retrieval quality could be improved by incorporating re-ranking or hybrid retrieval methods (e.g., combining sparse and dense signals) to refine final results. Finally, for scalability, it is important to design generalizable prompts and ER schemas, allowing KG-based HyDE to be easily adapted to new domains and knowledge bases without extensive manual tuning.

8 Limitations

8.1 Insight from the baseline reproduction and error analysis.

The error analysis revealed that HyDE’s effectiveness is not uniform across query types. The 15 queries where HyDE helped were shorter (mean 9.5 words vs. 12.5 overall) and lacked specialized scientific vocabulary, allowing the hypothetical document to act as an effective vocabulary bridge. Conversely, the 8 queries where HyDE hurt exhibited "embedding drift," in which the LLM-generated hypothetical document introduced tangential details that pushed the embedding away from the gold document. In all 8 cases the gold document remained in the top 100, suggesting the drift is moderate but sufficient to degrade precision at shallow cutoffs. The 25 both-miss cases involved highly specific quantitative claims or niche biomedical terms that defeated both methods, likely due to insufficient coverage in both the encoder’s training data and the LLM’s parametric knowledge. Finally, while HyDE improved Recall@10, it slightly lowered MRR@10 (69.4 → 69.0), indicating that HyDE does not always place retrieved documents higher in the ranked list and can sometimes push the gold document to a lower position than the baseline.

8.2 Capabilities needed for a better approach

The failure cases point to two key missing capabilities. First, the model needs the ability to constrain hypothetical generation so that it expands query vocabulary without drifting into tangential territory. Second, it needs awareness of when a query is already sufficiently specific that aggressive generation is unnecessary or even counterproductive, as reflected in the MRR@10 decline across the 252 both-hit queries.

8.3 Strength of the baseline

An important strength of vanilla HyDE is its strong vocabulary expansion for underspecified queries. The 15 HyDE-helped cases demonstrate the importance of keeping HyDE in our proposed approach to deal with shorter queries. Additionally, the 84% both-hit agreement rate shows that the dense encoder already handles most queries well with minimal overhead. Our approach should avoid adding too much complexity that degrades performance on this majority of already-working queries.

8.4 Refinement on the proposed approach

The error analysis provides direct empirical support for our KG-grounding approach. The 8 embedding drift cases are exactly the failure mode that KG grounding is designed to address. By constraining the LLM’s hypothetical generation with explicit entity-relations retrieved from a knowledge graph, we limit the room for tangential detail injection that caused rank degradation (e.g., Query 690 dropping from rank 2 to rank 60). Furthermore, the MRR@10 decline (69.4 $\rightarrow$ 69.0), driven by rank-shuffling within the both-hit set, suggests that even "successful" HyDE retrievals are subtly destabilized by unconstrained generation. This is another symptom that KG grounding can mitigate.

8.5 New ideas emerging from the error analysis

Beyond validating our core approaches, the error analysis suggests a complementary strategy of selective application. Since query length strongly predicts whether HyDE helps or hurts, a simple heuristic or classifier could decide per-query whether to apply hypothetical generation at all, bypassing it for already-specific queries while applying it for short, underspecified ones.

9 Ethical Considerations

Biomedical retrieval stakes. SciFact’s corpus is biomedical, where retrieval failures carry asymmetric costs: surfacing the wrong abstract on treatment efficacy or disease mechanisms can contribute to medical misinformation. KG-Grounded HyDE is a research artifact evaluated on a benchmark, not a system suitable for clinical decision support; deployment in such settings would require domain-expert validation and evaluation well beyond ranking metrics.

Inherited biases from LLM-built knowledge graphs. Unlike standard HyDE, our KG is constructed by prompting an LLM (GPT-4o-mini) to extract entity–relation pairs over the corpus. Whatever the LLM under-extracts or skews (under-coverage of non-Western medical traditions, uneven representation of rare diseases, inconsistent handling of contested findings) becomes a structural property of the retrieval substrate rather than a per-query artifact. This concentrates representational risk in an offline preprocessing step that is easy to overlook. Future work on KG-grounded retrieval should treat the extraction model as a first-class object of evaluation.

Reduced but not eliminated hallucination risk. KG-grounding reduces embedding drift but does not produce verified retrieval. Constrained hypotheticals can still contain incorrect entity–relation combinations when the KG is sparse, and a more fluent grounded hypothetical may earn more downstream trust than a clearly speculative one even when wrong. Our gains in MRR@10 and nDCG@10 measure ranking quality against gold annotations, not factual reliability.

Scope limitations. SciFact is English-only, abstract-only, and biomedical-only, with short fact-checkable claims. Real-world queries are often multilingual, multi-hop, or non-factual, and full papers contain methodology and caveats that abstracts elide. Our results should be read as evidence about a specific operating regime, not a general claim about retrieval quality.

Responsible build-on. Researchers extending this work should report extraction-model identity alongside retrieval results, evaluate across multiple domains and languages before claiming generality, and pair retrieval evaluation with downstream factuality evaluation when feeding a generation pipeline. KG construction also scales linearly in LLM calls with corpus size, a non-trivial cost at web scale that practitioners should weigh against retrieval gains.

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